

Enhancing ODIN's Predictive Accuracy with Six Data Feeds

Overview and Goals

ODIN is a biotech catalyst prediction system aiming for exceptionally high accuracy ($\geq 95\%$) and well-calibrated probabilities (Brier ≤ 0.054) on FDA decisions (PDUFAs) and Phase II/III trial outcomes. To achieve this, ODIN must leverage every bit of pre-catalyst information (strictly up to T-1, the day before an event) to foresee successes and failures. We will enhance ODIN's existing core signals by integrating six Modular Content Providers (MCPs): **FinBrain**, **ClinicalTrials.gov**, **PubMed**, **ChEMBL**, **bioRxiv**, and **Indeed**. Each source offers unique features that can improve ODIN's calibration and help address known failure modes – for example, “silent” FDA complete response letters (CRLs) with little public warning, marginal efficacy trial failures, or manufacturing blind spots that current signals miss.

Below we outline an implementation-focused strategy for each data feed, including specific sub-signals to extract, their expected impact on predictions, and how to weight and incorporate them. We emphasize maintaining ODIN's existing scoring framework while *augmenting it* with these new features for predictive gain. Any new data is utilized in a pre-catalyst manner (no leakage of post-event information), and we avoid introducing new MCPs or overhauling the entire architecture. The recommendations include necessary preprocessing steps and logic to ensure each signal is structured and integrated properly.

FinBrain – Financial News & Insider Sentiment Signals

FinBrain provides AI-driven financial datasets, notably daily **news sentiment scores** for tickers and aggregated **insider trading data** from SEC filings ¹ ². These can introduce market-derived features reflecting investor expectations and insider confidence. Key signal recommendations from FinBrain include:

- **Pre-Catalyst News Sentiment Score (with Trend):** For each stock, pull FinBrain's normalized daily sentiment score (range -1.0 bearish to +1.0 bullish) over the T-30 (or T-60) day window preceding the event ³. Use the **most recent sentiment** and **trend** (e.g. change over last 1-2 weeks) as features. A strongly negative sentiment heading into a catalyst may flag growing pessimism or rumor of trouble, whereas a strongly positive sentiment suggests optimism. We will interpret scores using FinBrain's definitions (e.g. < -0.5 = “strong bearish” ⁴) to trigger only when sentiment is extreme. The sentiment trend (slope/volatility) can capture sudden drops or spikes in mood that might precede news leaks. These features add a **market mood indicator** to ODIN's model. However, we will treat news sentiment as a *secondary* signal – useful for context but prone to noise – and calibrate its weight modestly ⁵. For FDA approval decisions in particular, news tone has historically shown weak predictive power ⁶, so we may even dial this signal down to near-zero for PDUFA events, focusing more on objective inputs. By contrast, for trial readouts (Phase II/III), speculative sentiment can sometimes foreshadow results (e.g. traders' chatter), so we'll allow a bit more influence there, with careful monitoring. Implementation-wise, we'll use FinBrain's API to fetch the

sentiment time series, convert string scores to float, and compute the latest value plus a short-term trend metric (e.g. 7-day change). These can be updated up to the evening of T-1. Any one-day anomalies will be smoothed or flagged to avoid overreaction.

- **Insider Trading Activity (Buys vs. Sells):** FinBrain's *Insider Transactions* feed collates Form 4 insider trades in near real-time ⁷ ⁸. We will derive an **"insider confidence" score** from the six months preceding the catalyst, focusing on *open-market buys* and *clustered sells*. Insider buying by top executives or directors is a rare but **extremely strong bullish signal** in biotech ⁹. ODIN's forensic analysis confirms that when insiders put significant personal money on the line pre-catalyst, positive outcomes are highly likely ⁹. Thus, our feature will capture the **net buy magnitude** (e.g. total dollar value of open-market purchases) and any **large individual buys** above a threshold (e.g. \$100K+, aligning with FinBrain's scanner examples ¹⁰ ¹¹). We will flag if a board director or CEO made an especially large purchase, as these carry even more weight (often termed the "board confidence" indicator) ¹². Conversely, insider selling is ubiquitous and often planned (10b5-1 programs, option exercises) ¹³, so we will mostly treat routine sells as neutral noise ¹⁴. The exception is a **cluster of unplanned sales** by multiple insiders ("cluster exit") or very large discretionary sales near the event – those can indicate insiders' pessimism and will be treated as a bearish signal ¹³. FinBrain's data includes transaction type labels (buy, sell, option exercise, etc.) and even identifies automatic 10b5-1 sales ¹⁴, which we'll exclude or down-weight as they are not informative by design. The output "insider confidence" feature could be a categorical flag (e.g. *Strong Buy Signal, Neutral, Caution* based on net activity) or a numeric score (scaled by total net shares or dollars bought minus sold). In integration, **insider signals will receive a hefty weight in ODIN's ensemble**, reflecting their outsized predictive value ¹⁵. For example, any **significant open-market insider buying will substantially boost the model's predicted success probability**, often enough to override minor negatives ⁹. Internal audit feedback explicitly recommended increasing the weight of insider sentiment in the model (up to ~25% contribution) while dialing back pure market momentum ¹⁶ ⁵. We will follow that guidance, ensuring insider buys meaningfully sway predictions. Likewise, a confirmed cluster sell-off will impose a notable penalty on the probability of success – though tempered if other factors (e.g. strong clinical data) are overwhelmingly positive.

Justification & Impact: These FinBrain-derived features introduce a view of **what those "in the know" are signaling**. Insider trading is an **objectively proven predictor** – corporate insiders' bullish actions have correlated strongly with positive trial/approval outcomes in case analyses ⁹, so capturing this will directly reduce ODIN's false-negatives on approvals (no more missing a quiet but confident success). It helps catch "silent" outcomes: e.g. an upcoming FDA decision with no obvious public signals might still be anticipated by insiders buying stock, tipping us off to approval despite silence elsewhere. On the flip side, if insiders quietly head for the exits (and it's not just pre-planned diversification), ODIN can flag a potential unseen problem (e.g. impending CRL or trial failure) that existing signals missed. News sentiment, while noisier, adds a **crowd sentiment check** – if, say, sentiment plunges inexplicably two days before a data readout, the model can modestly adjust downwards or at least prompt scrutiny, potentially catching issues like rumored FDA concerns or competitor news that could affect outcomes. We will apply interaction rules between these FinBrain signals as well: for instance, if **insider buys are strong but news sentiment is bearish**, we'll generally trust the insiders and not let media pessimism overly drag the prediction (as insiders have better information) ⁵. If sentiment is very negative and *no* insider buying is present, that confluence reinforces a cautious stance. The technical integration involves using FinBrain's REST API (or Python SDK) to fetch the necessary data nightly. We must ensure data cutoff at T-1 (no day-of data). Insider transactions will be aggregated and categorized (ignoring small sales, highlighting large buys). We'll implement logic per

FinBrain's guidance (e.g. treat purchases as bullish, 10b5-1 sales as neutral) ¹⁴ . By calibrating the weight of these features (heavy for insider buys, light for sentiment unless extreme) ¹⁵ ⁵ and segmenting by catalyst type (minimal sentiment influence on PDUFA's ⁶ , moderate on trials), we improve ODIN's calibration. This directly addresses failures where ODIN historically lacked insight – e.g. **recent audits showed ODIN underweighted insider info and overweighted news** ¹⁷ ⁵ , a gap we are now closing.

ClinicalTrials.gov – Trial Design & Progress Indicators

The **ClinicalTrials.gov** feed (via API) offers structured information about clinical study designs, status, and timelines for the trials underlying each catalyst. We will mine this registry data to extract **features about trial progress and design** that signal risk or confidence in outcomes. Key sub-signals from ClinicalTrials include:

- **Timeline Adherence and Delays:** Compare the trial's *original posted completion dates* to actual completion or last updated dates. A significant delay or extension in primary completion (beyond what was initially planned) can indicate enrollment difficulties or unforeseen issues. For each pivotal trial linked to the catalyst, we'll compute a "delay factor" (e.g. actual completion date minus planned date, or whether the trial required a protocol amendment to extend duration). Delays might signal trouble: for a Phase III readout, a delay could mean slower recruitment (perhaps due to lack of efficacy attracting patients) or additional follow-up requested, which could foreshadow marginal results. Conversely, finishing on schedule or early (without an announced early stop for futility) is a mild positive signal of smooth execution. We will incorporate a penalty for **major unplanned delays** pre-catalyst, particularly if accompanied by *no clear public explanation*. This helps identify cases where a trial silently struggled (which could lead to underpowered or inconclusive outcomes).
- **Single vs. Multiple Pivotal Trials:** Determine if the FDA submission or Phase III result is based on a **single pivotal trial or multiple trials**. This can be done by querying the registry for all Phase III trials for the drug/indication: if only one exists (or one plus perhaps a supportive Phase II), that event carries higher uncertainty. Historically, relying on one trial is riskier for approvals – FDA traditionally prefers two independent positive trials (the "two-trial paradigm") unless there's a compelling reason ¹⁸ . Our plan is to **flag single-trial approvals** as higher risk and adjust probability downward accordingly. The ODIN enhancement plan explicitly calls for checking if multiple trials were listed or just one, and factoring that in ¹⁹ . We will implement this by pulling the NCT records for the relevant trials and counting them. For example, if a PDUFA is supported by only one Phase III study, ODIN might apply an automatic modest haircut to the approval probability (unless it's an area like oncology or rare disease where single trials are more common – we can modulate the penalty by therapeutic area). Likewise for Phase III readouts: a program that has two concurrent Phase III studies (e.g. duplicate trials) gives more confidence that one success isn't a fluke, so a single-trial program will be treated with more caution.
- **Geographical Distribution of Sites:** Extract the fraction of trial sites that are in the US vs. international ¹⁹ . The rationale is that trials conducted predominantly overseas *might* face additional FDA scrutiny or data variability issues. A feature "% US sites" will be added. If a pivotal trial has a very low U.S. site percentage (or heavy reliance on regions with historically differing outcomes), we may slightly downgrade the approval probability, reflecting potential FDA concerns about generalizability or data quality. Conversely, a trial with a broad U.S. footprint (or North America/Europe concentration) might be considered more reliable. We will fetch the `location_countries` field via

the API to compute this. This “global trial” signal was suggested for implementation ¹⁹ and will help catch hidden risk factors – e.g. an all-China Phase III might carry CMC and data concerns for the FDA.

- **Endpoint Design and Changes:** Parse the trial's *primary endpoint and any protocol changes*. Surrogate or novel endpoints will be noted as higher risk (since regulatory agencies can be skeptical of surrogate endpoints translating to real benefit ²⁰). If ClinicalTrials.gov shows that the primary outcome measure is a surrogate (for instance, a biomarker level rather than clinical outcome), we tag a risk factor. Additionally, we'll check if the trial record had any **changes in endpoints or sample size** (which can sometimes be seen in version histories). A mid-trial change – e.g. switching the primary endpoint or increasing target enrollment – often means the original assumptions weren't panning out. For example, if an interim analysis led to adding more patients, that suggests the effect size might be smaller than expected (a warning sign for marginal efficacy). We will use the API or XML data to detect if `primary_outcome` descriptions changed or if `enrollment` was updated to a higher number. These will feed into a binary feature like “endpoint amended” or a continuous one like “enrollment increase factor”. Such signals directly address **marginal efficacy failures**: a trial needing adjustments to hit significance is more likely to yield borderline or negative results.
- **Adverse Event Signals (if available):** In some cases, ClinicalTrials.gov records may contain reported serious adverse events or reasons for termination (if a trial terminated early). While not common for still-ongoing trials, if a trial ended pre-maturely or had an *expanded access* program, we can glean safety signals. We will scan for any mention of serious adverse events in the trial results (if posted before T-1) or any record that the trial was halted due to safety. This is likely rare (and major safety issues would be press-released), so this is a minor, opportunistic signal. It's mentioned as a possible data point to mine in an advanced safety analysis ²¹, but we consider it optional/bonus due to low availability before catalyst events.

Justification & Impact: Incorporating trial design/progress features gives ODIN a **deeper understanding of the context behind each catalyst**, beyond what press releases alone provide. These signals improve prediction calibration by adjusting for structural risk factors in the trial: e.g. a drug filing based on a single, mostly overseas trial with a surrogate endpoint should have a lower predicted success *even if* prior data looked good, because historically such situations often lead to CRLs or requests for another study. By penalizing these features, we aim to reduce false positives (overestimation of success when the setup is shaky). Notably, many “silent CRL” cases in the past can be retrospectively explained by such factors (for example, FDA quietly worrying there's only one trial or manufacturing done abroad). Now ODIN will catch those. Conversely, when a program has two well-run, on-schedule Phase IIIs with hard endpoints, ODIN can be more confident (raising probability of success appropriately). These adjustments address **calibration** per segment: as recommended, regulatory approval predictions should incorporate factors like trial count and modality ¹⁹ ²² so that a 70% predicted probability means something different in a Phase II context versus a PDUFA ²³. In practice, we will parse trial data via the ClinicalTrials.gov API for each catalyst event (mapping the drug or indication to its NCT ID). A preprocessing script will pull fields (status, dates, enrollment, locations, outcomes) and produce structured features. We will integrate these into ODIN's model or rule engine. For example, we might implement a rule: “If PDUFA and only 1 pivotal trial, multiply base probability by 0.9” (a mild cut) unless certain exceptions apply (like oncology orphan). Or use a learned coefficient in a regression model that naturally lowers odds for single-trial entries. Similarly, a large delay might subtract a fixed amount from the confidence score. We will refine the magnitude of these penalties using historical analysis – e.g., check past cases where delays or single-trial submissions led to failure, to calibrate how strongly to weight these features. This ensures we **reduce Brier score** by aligning predictions

with real-world odds (for instance, eliminating the optimism bias on single-trial approvals). The enhancement plan explicitly notes that adding such precedent signals (like trial count) should improve Phase III prediction discrimination ²⁴, which aligns with our goal of catching marginal efficacy issues. Overall, ClinicalTrials data acts as a **“reality check” layer**, grounding ODIN in the operational and design realities that portend success or failure.

PubMed – Scientific Literature Insights

PubMed provides access to peer-reviewed biomedical literature, which we will tap for **qualitative signals about the drug, target, or indication**. The idea is to quantify the scientific context: how strongly does independent research support the mechanism? Are there red flags in publications that haven’t made it to press releases? Key sub-signals to derive from PubMed include:

- **Literature Volume and Trajectory:** Measure the number of publications (and review articles) in recent years related to the drug’s target, mechanism, or the specific therapeutic approach. A rich and growing literature indicates that the science is well-explored and validated by multiple groups, whereas a sparse literature suggests a novel or unvalidated mechanism. We will use PubMed queries (via NCBI APIs) for the target gene/protein, pathway, or the drug molecule name (especially if the drug has a known INN or research code used in publications). For example, if the catalyst is a new IL-12 inhibitor, we’d check how many papers in the last 2–3 years discuss IL-12 inhibition in the disease area. We will create a feature for **publication count trend** (e.g. publications in last 1 year vs. prior year). An increasing trend can mean rising scientific enthusiasm (or concern) about that approach. This can improve calibration: a therapy backed by a broad body of evidence might deserve a higher prior probability of success, whereas one with scant external research could be more speculative (thus slightly lower baseline). It’s a form of measuring **scientific consensus**.
- **Independent Validation or Skepticism:** Look for specific content in the literature that could influence outcomes – for instance, **papers that replicate or refute earlier stage results** of the drug or similar compounds. If a Phase II result was published, have other scientists cited it positively or have follow-up studies confirmed the findings? Conversely, are there editorials or studies pointing out flaws or risks? We can do targeted searches for the drug name or class along with keywords like “efficacy”, “toxicity”, “mechanism” etc. If we find, say, a recent *Journal of Medicine* article where experts question whether the drug’s surrogate endpoint will translate to clinical benefit, that’s a qualitative negative signal. Or if multiple papers report that drugs of this class show, for example, off-target toxicity in animal models, we should flag that. While it’s challenging to quantify this automatically, we can at least set up a **“literature sentiment scan”**: using text mining on abstracts to detect positive vs. negative tone about the approach. Alternatively, simpler proxies like the presence of **high-profile publications** can serve – e.g. if the Phase I/II results were notable enough to appear in a top journal (NEJM, Lancet) and get cited widely, it suggests robustness. We will incorporate binary flags for any *notable publication milestones* (published in top journal, or highly cited paper on the drug). A highly cited paper implies strong interest; indeed, “citation velocity” (the rapid accumulation of citations) is used in bibliometrics to identify important findings ²⁵. We interpret a high citation velocity of relevant papers as the field finding the results useful and credible ²⁶. That would support a positive outlook. In contrast, if the drug’s earlier studies never got published or have very low citation, it might indicate lack of interest or hidden issues.

- **Safety and Mechanism Red Flags in Literature:** Often, certain safety concerns or mechanistic shortcomings surface in scientific papers before they are widely recognized. We will search for any **case studies or preclinical research** hinting at problems – for example, papers about the target’s role in an unintended pathway, or animal studies reporting toxicity. If PubMed yields an article like “Drug X, a [class], induces liver toxicity in long-term rodent studies,” that’s a clear warning sign to incorporate. Another example: if the target is part of a pathway with a known feedback loop, a pharmacology journal might have published that blocking it causes some counteracting effect (potentially reducing efficacy). We plan to capture such insights by keyword scanning relevant to the program (each catalyst will have a brief biologic description we can use to formulate searches). Detected red flags will translate to features such as **“literature safety flag = true”** or **“efficacy mechanism controversy = true”**. These would directly penalize the prediction if present.

Justification & Impact: PubMed-derived signals ensure ODIN’s predictions align with the *broader scientific reality*. They function as a **calibration layer for unknown unknowns**: for instance, ODIN’s base model might give a high probability to a Phase II trial if prior company data was good, but if the literature reveals that virtually every attempt at this mechanism has failed or that experts are skeptical, ODIN should temper its confidence. Incorporating a literature consensus score can reduce overconfidence in cases prone to failure (thus lowering Brier score by avoiding predictably wrong high probabilities). Likewise, literature signals can help catch **marginal efficacy issues**: e.g. if multiple independent studies suggest only a mild benefit of this mechanism, ODIN can be biased a bit towards expecting a negative or modest outcome even if the company is optimistic. For “silent” failures or CRLs, literature might occasionally tip us off – e.g. a toxicology study by an academic might highlight a safety issue that wasn’t publicized by the company. If ODIN sees that, it can anticipate a possible safety-driven CRL. In terms of interactions: if literature sentiment is strongly at odds with insider or market sentiment, we’ll have to balance them. For example, insiders might be buying (optimistic), but suppose a very recent publication by independent researchers questions the drug’s fundamental premise. We might neither fully discount the insider signal (they have data) nor ignore the scientific critique. The model could land in a middle-ground probability (or flag the uncertainty). Overall, these signals likely won’t swing predictions as dramatically as clinical or insider data, but they **fine-tune calibration** and catch subtler risks. Implementation-wise, we’ll set up automated PubMed queries triggered for each event (using known drug name, target name, and indication keywords). A preprocessing script can retrieve counts (to get volume metrics) and sample abstracts for NLP analysis. We might leverage sentiment analysis or simple heuristic rules (e.g. count occurrences of words like “significant improvement” vs “failed to” in abstracts). Additionally, citation counts for key papers can be pulled from tools like Semantic Scholar or CrossRef to compute “citation velocity” – for instance, if a paper from last year already has dozens of citations, that’s noteworthy ²⁶. These data points will then be fed into ODIN’s feature set. Some signals (like a specific safety flag) might be used in a rules-based manner (e.g. if an academic paper raises a serious concern, explicitly knock down the prediction). We will document any such rule for transparency. By integrating PubMed insights, we ensure ODIN’s predictions are not made in a vacuum but rather cross-validated against the collective knowledge of the scientific community.

ChEMBL – Molecular Properties & Toxicity Signals

ChEMBL is a comprehensive database of drug molecules, targets, and bioactivity data. We will use ChEMBL to extract **molecular-level features** of the therapeutic candidate that could indicate development risks –

especially latent safety issues or formulation/manufacturing challenges. Key signals to derive from ChEMBL include:

- **Off-Target Pharmacology & Toxicity Indicators:** Query ChEMBL for any bioassay results or known activities of the drug (or close analogs) on safety-relevant targets. For example, a critical flag is **hERG channel binding affinity** – if the molecule shows sub-micromolar potency for the hERG potassium channel (K_v11.1), it suggests a risk of QT prolongation and cardiac toxicity. We will retrieve any ChEMBL assay entries for the compound against hERG or other cardiac ion channels. A feature like “*hERG_inhibition = True*” would be set if, say, $IC_{50} < 1 \mu M$ ²⁷. Similarly, we’ll look for assays on CYP enzymes (to gauge drug-drug interaction liability) or known toxicity pathways (like Ames test for mutagenicity, if available). If ChEMBL indicates the compound was flagged in a **toxicity assay (e.g. positive in a hepatotoxicity screen or genotoxicity test)**, we will incorporate that as a strong negative predictor. Many CRLs and phase failures stem from safety findings that might have been predictable from chemical properties – this feature helps ODIN anticipate those. For example, if a small molecule has a structural alert for reactive metabolites (and ChEMBL data or literature shows it causes liver enzyme elevations in preclinical tests), ODIN can pre-emptively assign a lower probability of success (since such issues often surface in Phase III or in FDA review).
- **Structural Alerts & Drug-Likeness:** Compute or retrieve from ChEMBL certain molecular descriptors that correlate with developability. ChEMBL’s molecule records include properties like molecular weight, LogP, and structural fragments. We can define rules for known **toxicophores** (toxic substructures) – e.g. presence of a nitro-aromatic group, anilines, etc., which medicinal chemists consider red flags. If the compound contains any such substructure, flag a potential risk. Additionally, extremely high lipophilicity (LogP) or very high molecular weight could indicate formulation difficulties or off-target binding promiscuity, which might lead to safety or CMC problems. We will pull the compound’s properties via ChEMBL API (`/molecule` endpoint) to get these. For instance, a $LogP > 5$ or molecular weight > 800 might be out of the drug-likeness range, warranting a small penalty as such compounds often encounter delivery or stability issues. This is a softer signal, but it contributes to ODIN’s assessment of how *chemically tractable* the drug is. A saying in pharma is “the process is the product” ²⁸ – if a molecule is very complex, manufacturing and consistency can be a nightmare (leading to CMC CRLs). Thus, a “**formulation complexity score**” could be derived (perhaps from number of chiral centers, compound class, etc.). We can use ChEMBL’s compound class info (small molecule, peptide, etc.) combined with rules (e.g. large synthetic peptides or toxins are harder to manufacture) to inform this.
- **Target Class and Similar Compound Outcomes:** Leverage ChEMBL’s target annotations to identify the drug’s biological target(s) and see how other compounds against that target have fared. If the target is one that has seen many failures historically, we should adjust for that. For instance, if the drug targets an **amyloid pathway enzyme** (and we know many past attempts failed), that’s a risk factor. ChEMBL can list known ligands for the target; we might cross-reference which of those became approved drugs vs. were terminated. While a full analysis might go beyond ChEMBL, even a qualitative indicator can be used – e.g. *target_novelty = high* (no approved drugs for this target yet) vs. *target_novelty = low* (well-trodden target with successful drugs). A high novelty means more uncertainty (thus slightly lower predicted success due to unproven biology). Conversely, a target that already has approved drugs in other indications might have a higher chance of success. If available, we might find in ChEMBL or literature any **close analogs** of the current drug and note their fate (if

an analog failed in Phase III for toxicity, that's a cautionary tale). This signal overlaps with PubMed/literature research but here we use ChEMBL's structured data to systematically flag it.

Justification & Impact: ChEMBL features equip ODIN to catch **“under-the-hood” risks that aren't evident from clinical data alone**. A substantial fraction of late-stage failures and CRLs come from safety or CMC problems – for example, an unexpected toxic metabolite or difficulty reproducing the product at scale ²⁷ ²⁹. By identifying molecular toxicity risks (like hERG binding), ODIN can predict safety-related failures more often instead of being blindsided. This should markedly improve recall of failures – our goal is to boost CRL (failure) recall above 50% from a low baseline ³⁰ ³¹, and catching hidden CMC/tox issues is crucial to that. For instance, if a PDUFA drug has no clinical red flags but our ChEMBL analysis shows a potent hERG inhibition, ODIN might assign a “hidden risk” penalty, potentially converting what would have been a false prediction of approval into a more cautious prediction (aligning with a possible CRL due to QT concerns). This directly reduces over-confident predictions and hence lowers Brier score. Another scenario: if a Phase II trial barely met its endpoint, and we see the compound is extremely large and hard to formulate, ODIN might foresee that it won't progress smoothly to Phase III or approval (maybe due to dosing issues), thus predicting failure where a naive model might not.

Integration-wise, we will set up a pipeline to query ChEMBL by compound name or structure. The **preprocessing** will fetch the compound's ChEMBL ID (if available) and gather: known activities on a predefined panel of safety targets (hERG, key CYPs, etc.), basic physicochemical properties, and target information. We may also use ChEMBL's similarity search to find close compounds and check if any notes on their development exist (though this might require manual curation or an internal knowledgebase in addition). We'll codify simple rules: e.g. *if hERG_IC50 < threshold then safety_flag += 1*. These can be combined into a **molecular risk score**. In ODIN's model, this could be an input to a specialized “CMC/Toxicity risk sub-model” – essentially acting as a classifier that outputs a high risk probability if multiple chem flags are tripped. In fact, the system enhancement plan suggested a dedicated *CMC/Inspection Risk oracle* module ³². Our ChEMBL-based features would feed that module, alongside any other manufacturing clues. We will weight these signals strongly when clear-cut (e.g. an Official Action Indicated (OAI) at a manufacturing plant, if known, would almost single-handedly flip to AVOID ³³; we mirror that philosophy for severe chem tox flags – they should have a large negative weight). For milder signals (like high LogP), we assign smaller weight, just nudging predictions. We will also incorporate **modality checks**: if ChEMBL indicates the therapy is not a small molecule (e.g. a biologic or gene therapy isn't well-represented in ChEMBL data), we fall back to a general *modality complexity factor*. As noted, advanced modalities like gene therapies should be penalized more due to higher manufacturing and regulatory risk ²². For example, if ODIN identifies the product as a gene therapy (perhaps via its mechanism keyword or trial description), we apply a multiplier that lowers success probability unless mitigating factors exist. This catches manufacturing blind spots because many “silent” CRLs have been for gene therapies or complex biologics where CMC issues emerge late. In summary, ChEMBL signals harden ODIN against hidden chemical liabilities, ensuring our predictions aren't overly optimistic just because clinical data looked fine. By quantifying molecular risk, we aim to prevent surprises where ODIN would have given a 90% approval chance but a known chem issue causes a CRL – instead, ODIN will have factored that in and perhaps only given 60-70%, better aligning with reality (hence improving calibration).

bioRxiv – Preprint Research Trends

bioRxiv (and its sister site medRxiv) host preprint papers, which are early scientific reports not yet formally published. They represent the *bleeding edge* of research. We will utilize bioRxiv to gauge **emerging**

scientific discussions and data around the drug or its mechanism, especially in the months leading up to the catalyst. Key signals to gather from bioRxiv include:

- **Emergence of Relevant Preprints & Citation Velocity:** Monitor bioRxiv for any new preprints mentioning the drug, target, or disease in question in the lead-up to the catalyst. If, for example, a prominent research group posts a **preprint about the drug's mechanism** (or a competing approach) shortly before the event, this could be significant. We will create a feature that counts the number of *relevant preprints in the last ~6 months*. Beyond just the count, we will track **attention metrics** for those preprints – namely, how quickly they are being accessed or cited (citation velocity). Citation velocity is essentially the rate at which a paper accumulates citations or mentions after release ²⁵, and it helps identify “hot” papers that the community finds important ²⁶. In practice, for each relevant preprint, we can use altmetric indicators (like number of Twitter mentions, or if it's been cited by other preprints or even PubMed articles within a short time). A high citation velocity indicates that the preprint's findings are resonating in the field. If such a preprint contains *positive* findings supporting the drug's mechanism (e.g. an academic study that confirms a biomarker improvement with this approach in an animal model), and it's gaining attention, that bolsters confidence in the drug's success. Conversely, if a preprint reports a **negative or contradictory result** – say, a new experiment where the drug's target was knocked out and produced no improvement in disease – and that preprint is circulating widely, it could foreshadow problems for the drug (the scientific community picking up on a potential flaw). We will incorporate a **“preprint buzz score”** combining the count of new preprints and their attention. A spike in preprint activity could modestly adjust ODIN's predictions: spikes around *positive* developments might increase our predicted probability of success (since it implies validation), whereas spikes around *negative* findings would decrease it. If there's no preprint activity at all, that's the baseline (no adjustment).
- **Qualitative Preprint Content Flags:** We will also attempt to parse the content of any identified preprints for specifically relevant information. Preprints often contain detailed data or hypotheses that haven't made it to formal publication yet. For example, a preprint might reveal **mechanistic details**: “Compound X was found to induce off-target immune responses” or “New trial of a similar drug Y failed to show benefit.” These tidbits can be turned into flags for ODIN. We will scan titles/abstracts for phrases like “*failed to improve*”, “*not significant*”, “*toxicity*” or on the flip side “*robust improvement*”, “*significant efficacy*”. Based on that, if a preprint's conclusion is clearly negative for the class of drug, we set a negative flag. If clearly positive or a breakthrough supporting evidence, a positive flag. The model can use these as additional evidence. For instance, a **silent CRL** risk might be detected if a preprint by independent researchers demonstrates a safety concern that was previously unknown – even if the company hasn't addressed it, the FDA might know and be concerned too. Such a scenario is rare but exactly the kind of subtlety these signals cover. Another use: if a company's own scientists publish a preprint about some aspect of the drug (like maybe pharmacokinetics in a subpopulation) right before the readout, it might indicate something (either they are trying to drum up support, or explain something). ODIN will monitor for these moves.
- **Competitive Landscape Monitoring:** bioRxiv can also alert us to **competitor results**. If a competing molecule in the same class releases data (even as a preprint or conference abstract posted on a preprint server) showing either a big success or failure, that indirectly impacts our drug. For example, if a competitor's Phase II (in the same mechanism) flops and a scientist posts an analysis on medRxiv, it could mean trouble for our drug as well – class effect failures. ODIN can factor that in as a negative signal for our drug's Phase III. Conversely, if a competitor shows a strong

positive result, it might indicate the mechanism works (positive class validation) – boosting confidence in our drug (though one could argue a competitor success might threaten market share, here we focus on technical success probability, not market outcome). In any case, competitor preprint signals need to be incorporated carefully; we'll likely use them qualitatively with human review for now (to ensure we interpret correctly), but the presence of any competitor data in preprints is a useful feature by itself (it means the field is active and outcomes are being revealed).

Justification & Impact: bioRxiv signals complement PubMed by catching the **most up-to-date scientific developments**, often ahead of peer review. This is crucial for staying *ahead of the curve* – ODIN can react to information that others might not formally publish for months or ever. While not every catalyst will have relevant preprints, when they do appear, they can be game-changers for prediction. For instance, consider a marginal efficacy failure scenario: ODIN's base model might be 70% confident in success, but a week before the readout, a preprint from a respected lab reports that the biomarker the drug targets didn't correlate with clinical outcome in a long-term study. This directly challenges the drug's rationale. Our pipeline would catch this preprint, flag a negative signal, and ODIN could smartly dial down that 70% to, say, 50% or less – possibly saving us from an overprediction and improving Brier score on that event. Similarly, for a silent CRL: perhaps a preprint discloses that the manufacturing process yields an impurity – something that might cause a CRL. If by luck that's posted publicly, we capture it and reduce our approval prediction accordingly. These are edge cases, but when they occur, they significantly affect outcomes, and having the capability to detect them keeps ODIN's accuracy high. Even in more common situations, general “buzz” around a therapy can be informative. A high volume of preprints and citations implies a lot of scientific momentum – which often correlates with true innovation (and successful results) or at least greater scrutiny (which could uncover problems). Either way, ODIN benefits by not ignoring this dimension.

From an implementation standpoint, we will periodically run searches on bioRxiv's API or RSS feeds for the company name, drug name, target, and disease keywords. We'll restrict to times before the event. Any hits will be pulled and stored. We will use altmetrics services (if available via an API) or simply track citation count via Crossref for those preprints to compute velocity. Because preprints can be numerous and not all relevant, we might integrate a simple ML classifier to filter relevance (e.g. is this preprint actually about our drug's mechanism or just tangential). The outcome will be a small set of signals (counts, and any flagged sentiment). These will feed into ODIN's model likely with a low initial weight (since historical data to train on preprint signals is limited – this is mostly a forward-looking enhancement). As such, we may treat them as advisory signals: if strong evidence appears, we adjust predictions, but otherwise they don't interfere. We'll also log these instances for future validation (to build up a knowledge base of “preprint signals that mattered”). In summary, bioRxiv integration makes ODIN more **responsive and up-to-date**, potentially catching last-minute signals and thereby reducing prediction errors in edge cases. It's another step to ensure our model's knowledge cut-off is right up to T-1 in a meaningful way.

Indeed – Company Hiring Trends and Workforce Signals

Indeed job postings data will be harnessed to quantify the company's **hiring momentum and focus** leading up to the catalyst. The premise is that companies often adjust hiring (either ramping up or freezing) based on their internal expectations of success or failure. Indeed provides a window into these strategic moves. Key signals to extract from Indeed include:

- **Hiring Ramp Slope:** Calculate the rate of change in the number of open job listings at the company in the months before the catalyst. This involves scraping Indeed (or using their API if available) for all

jobs at the company (or relevant subsidiary) and tracking the total count over time (e.g. each month or bi-weekly). We then perform a linear fit or percentage growth calculation to get a **hiring slope (% increase per month)**. A positive, steep slope means the company is in a **hiring surge**. Just before a major catalyst, a hiring surge often indicates confidence and preparation – for instance, scaling up commercial teams, medical science liaisons, manufacturing staff, etc., anticipating a positive outcome. A negative slope (declining openings) or hiring freeze could indicate caution or belt-tightening – possibly because management is bracing for a failure or a delay. ODIN will use the hiring slope as a continuous feature: a strongly positive slope might contribute a small boost to the success probability (or at least counterbalance other negatives), whereas a zero or negative slope might trigger a slight downgrade in confidence. It's important to calibrate this by company size: a big pharma might always be hiring, so we'll likely focus this signal on small- to mid-cap biotech where catalyst outcomes would significantly impact hiring plans. For those, a distinct change in hiring trend is telling. If available, we'll also incorporate the **absolute number of new job postings** in key functions (see below) as another measure of ramp intensity.

- **Role-Specific Hiring Indicators:** Delve into the types of roles being advertised. Specifically, look for roles that align with **product launch or late-stage development needs**: for example, **commercial roles** (sales reps, marketing managers, market access) and **manufacturing/quality roles** (process engineers, quality control specialists, supply chain managers). An uptick in postings for commercial positions in the quarter leading up to a PDUFA is a classic sign that the company expects approval and is gearing up to launch. If we detect, say, 20 new sales positions listed in the last 3 months, that's a strong positive signal. In contrast, if a company with an upcoming drug launch is *not* hiring any commercial staff and hasn't announced a partner, it suggests they might not anticipate an approval (or at least not an immediate launch) – a potential red flag. We will create features such as **"commercial hiring count"** and **"manufacturing hiring count"** in the pre-catalyst period. Likewise for clinical development roles before a Phase II/III readout: are they hiring clinical trial managers or regulatory affairs specialists (implying they expect to move to next phase or file with FDA)? If yes, likely they are optimistic about the results. If not, maybe they are unsure or waiting. One specific sub-signal is the presence of **MSLs (Medical Science Liaisons)** or other field medical roles postings pre-approval – these roles are often brought on 3-6 months before an expected approval to educate physicians, and seeing a spike in MSL hiring is a bullish indicator that was noted for case studies ³⁴. We will definitely flag any such hiring burst. On the other side, scanning job titles can reveal if the company is trying to address a **problem area** – e.g. suddenly recruiting multiple "Manufacturing Compliance" or "CMC Project Manager" roles could imply they encountered manufacturing issues that require new expertise (this could mean they're scrambling to fix something the FDA flagged, which is a bad sign if it's right before approval). We'll interpret multiple new CMC/QA job postings pre-PDUFA as a **manufacturing risk signal**, ironically: it shows awareness of a gap, but so late that it might result in a CRL anyway.

- **Hiring Freeze or Retrenchment Signals:** In addition to postings, any evidence of job cuts or a drop in hiring can be telling. If our Indeed scraping shows a significant **decrease** in open positions (or if we see that many job listings were closed without being filled), it may indicate the company pulled back. Possibly, management might freeze hiring pending the catalyst outcome to conserve cash (which could mean they are not confident enough to invest ahead). We might infer this indirectly: e.g. if in earlier months they had many listings but by T-1 far fewer, without hires being made (just cancellations), that's a cautious stance. This signal would be a mild negative factor. Additionally, scanning news (via FinBrain's news feed) for words like "layoff" or "restructuring" at the company

before the catalyst is another angle – but since we limit to the given MCPs, we'll stick to Indeed data for direct info and rely on FinBrain's sentiment to possibly capture any layoff news.

Justification & Impact: Hiring data is a **proxy for insider expectations and operational readiness**. It directly addresses some failure modes, especially the **manufacturing preparedness and "silent" CRL risk**. For example, one known pattern is that many CRLs (especially for manufacturing) happen to companies that were not fully prepared to launch – either they lacked an established manufacturing team or plan (as seen in cases like Ocugen, which had unclear manufacturing and rightly was flagged as high risk) ²⁹. By examining hiring, we can detect that lack of preparation. If a company has an upcoming PDUFA but **zero hiring in manufacturing or commercial**, and no announced partner, ODIN will treat that as a major risk flag (the model might output a "CRITICAL risk – no launch prep" warning akin to the Ocugen example ²⁸). This will help catch those silent CRLs where the missing piece was on the CMC side – effectively reducing false positives (predicting approval when in reality the company wasn't ready). On the flip side, a company actively **hiring dozens of people in anticipation of approval** is unlikely to be doing so if they expect failure – this can give us confidence to increase the predicted success probability. It improves sensitivity to successes that might otherwise look risky on paper. For marginal efficacy scenarios, hiring signals also play a role: if the company believes in the drug, they might start the next trial prep, but if they secretly doubt the Phase II, they might hold off on hiring for Phase III. Thus, a lack of expansion in R&D roles could foreshadow a negative result.

We will integrate these signals with interaction logic. A prime example: if FinBrain's news analysis shows *no partnership deal* for commercialization and our Indeed data shows *no hiring of a salesforce*, that combination is a red flag for a PDUFA (because either they are extremely confident of partnering last-minute, or they suspect a CRL such that building a salesforce would be premature). ODIN would then heavily penalize the approval probability. Alternatively, if the company stated they will partner upon approval (news can sometimes indicate that), we adjust the expectation – in that case, lack of hiring is not a negative signal (so we'd have an interaction rule: "if known commercialization partner exists, ignore absence of sales hiring"). This ensures we interpret hiring data in context. Another interaction: tie hiring with insider trading – if insiders are buying *and* the company is hiring aggressively, all signs align to a high confidence in success (model should probably predict success unless glaring issues elsewhere). If insiders are selling but the company is still hiring, that's a conflicting signal (maybe the selling was planned or for other reasons) – we might then rely more on the objective hiring data for operational readiness but remain cautious.

From an implementation perspective, setting up the Indeed feed involves **scraping or using their API regularly (weekly or monthly)** for each company ticker. We'll maintain an internal time series of open job counts. We also need a mapping from ticker to company name as used in job listings (for example, some biotech have multiple subsidiaries or might list under a parent company name). We might use the company's LinkedIn or official careers page as cross-validation if Indeed data is sparse (many small biotechs post jobs on their own site). However, since our focus is Indeed, we'll primarily stick to that source. We'll parse job titles and descriptions for keywords: e.g. store a list of postings and classify each by function (Sales/Marketing, Manufacturing/Quality, R&D/Clinical, G&A). A simple keyword dictionary (e.g. "sales", "representative", "MSL" for commercial; "manufacturing", "engineer", "process", "GMP" for CMC; etc.) will allow tagging. We can then count per category. This structured dataset (counts per category, trend) will feed into the model. Initially, because we cannot backtest this easily on past data (lack of historical snapshots) ³⁵, we will treat these signals as qualitative adjustments with conservative weights. The audit noted that such signals may need forward-looking validation ³⁵, so we'll incorporate them in a way that they can only improve decisions if they are clearly present, and won't hurt calibration if absent. For instance, we might

say: by default, assume neutral; only apply a boost if a clear hiring surge in relevant areas is detected, and apply a penalty if a clear absence where there should be presence is detected. As we gather future examples, we'll adjust the magnitude. Already, anecdotal case analysis suggests this is valuable – e.g., companies that hired many MSLs pre-launch generally succeeded ³⁴, which backs our use of that signal.

In summary, Indeed-based signals enhance ODIN by adding a **human behavior dimension**: corporate actions often speak louder than press releases. By watching those actions (hiring), ODIN becomes more adept at reading between the lines and anticipating outcomes accordingly. This directly targets current failure modes like unseen manufacturing issues (no hires means no capacity, likely CRL) and improves overall confidence calibration for each prediction.

Implementation & Calibration Summary

Collectively, these six feeds will be integrated into ODIN in a modular fashion, each contributing to a specific aspect of the risk assessment:

- **FinBrain** provides **investor sentiment and insider trading signals**, which we'll weight heavily for insider buys (strong success predictor) and use carefully for news sentiment (small weight, mostly for extreme cases) ¹⁵ ⁵. This helps align the model with informed insider knowledge and avoid being misled by media hype or gloom, improving both recall of successes and avoidance of false optimism ⁹ ¹³.
- **ClinicalTrials.gov** adds **hard data on trial execution and design** (e.g. multiple trials or not, delays, endpoint types) ¹⁹. These features let ODIN adjust probabilities for structural factors (like single-trial filings, predominantly ex-US data, etc.) that frequently influence FDA outcomes. By doing so, we aim to catch more failures that result from trial design inadequacies and ensure our probabilities reflect realistic odds given the trial context.
- **PubMed** brings in **scientific consensus and external validation signals**. Using literature volume, independent findings, and expert viewpoints, ODIN can gauge whether a program is on solid scientific footing or not. This reduces overconfidence in scientifically tenuous programs and increases confidence (slightly) in well-validated ones, addressing subtle efficacy or safety pitfalls noted in papers.
- **ChEMBL** contributes **molecular and CMC risk factors**, directly targeting safety and manufacturability concerns. By flagging toxicological signals (like hERG) and complexity issues, ODIN will not treat all drugs equally, but rather penalize those that carry a higher inherent risk of late-stage failure ²⁷ ²⁹. This dramatically helps with capturing silent CRLs (many of which are due to CMC/safety) – our model will no longer ignore a looming 44%-of-CRLs cause ²⁹.
- **bioRxiv** offers **real-time research buzz** detection. It's a safety net for any last-minute or under-the-radar developments. While its impact might be case-specific, having it ensures ODIN is as informed as any expert scanning new abstracts, thus preventing misses in scenarios where everyone in the lab circles knows something that the market might not yet fully digest.

- **Indeed** gives **behavioral/operational signals** from the company itself. By reading the company's preparation or lack thereof, ODIN gains a quasi-“insider” perspective (short of direct insider trading) on how likely the outcome is perceived internally. This is especially potent for detecting readiness (or the lack of it) around manufacturing and launch – often the Achilles’ heel in biotech predictions ²⁹.

Each of these sources will be fed through appropriate preprocessing pipelines (APIs, scrapers, text mining as described) and then into ODIN's model pipeline. We plan to implement some as additional features in the existing logistic regression/ensemble model, and others as separate sub-models or rule-based overlays (per the design of ODIN v8.x which allows expert modules) ³⁶. Crucially, **calibration** will be maintained by segmenting models and weighting signals according to reliability. Where needed, we will apply segment-specific calibration curves so that, for example, PDUFA probabilities remain properly high on average (given their high base rate) even after adding new risk signals ²³. The new signals themselves will be calibrated via backtesting on whatever historical data is available (we may simulate or use partial data for ones like hiring that lack full history, doing case studies as needed ³⁴). The integration will also respect interactions – for instance, not double-penalizing if two signals flag essentially the same underlying issue (e.g. ChEMBL toxicophore and a PubMed paper on toxicity should together count as one strong issue, not two independent ones). We will code logical modifiers to handle that.

In conclusion, by **leveraging FinBrain, ClinicalTrials, PubMed, ChEMBL, bioRxiv, and Indeed feeds in concert**, we expect to significantly enhance ODIN's predictive performance. These sources arm ODIN with a 360° view: from **Wall Street sentiment to lab bench findings to factory floor hiring**. The model will be better at identifying high-risk scenarios (thus appropriately lowering probabilities and avoiding surprise misses) and at recognizing genuinely derisked situations (allowing high confidence calls). This should yield improved accuracy and Brier score, and particularly address prior failure modes like unanticipated CRLs (which we now target with multiple signals) and marginal efficacy misses. We will monitor the impact of each signal on validation sets and tune their contributions to ensure we hit the desired calibration targets (for example, aiming for CRL recall >70% with minimal false alarms) ³⁷ ³⁸. By constraining ourselves to pre-event data and these six feeds, we maintain ODIN's real-time applicability while greatly enriching its knowledge. The result will be a more **robust, informed, and reliable** prediction system for biotech catalysts, achieved through careful implementation of the above signal-specific strategies.

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